

Independent Replication Report:
Single Quantum Querying of a Database
Terhal & Smolin, arXiv:quant-ph/9705041 (Phys. Rev. A **58**, 1822,
1998)

QC-200 Replication Wave, 2026-07-05

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1 Paper Summary

Terhal and Smolin (1997) revisit Bernstein and Vazirani’s parity problem and package it as a class of *single-query* quantum algorithms for database-retrieval problems. The setting is a database Y encoded as an n -bit string y , queried through the oracle

$$R_y : |x, b\rangle \mapsto |x, (b + x \cdot y) \bmod A\rangle, \quad x \cdot y \equiv \sum_i x_i y_i \bmod A. \quad (1)$$

Classically, extracting the full n -bit string y requires at least $M \geq n/\log_2(n+1)$ oracle calls (Eq. (2.1) in the paper, from the standard information-theoretic bound Eq. (1.2)). The central claim is:

A single quantum query, sandwiched between Hadamard transforms on the query register and with the answer register prepared in an equal superposition with alternating signs, retrieves the entire n -bit string y with probability 1.

This is Bernstein–Vazirani’s parity algorithm applied to *coin-weighing* (binary answers, Sec. II) and generalised to binary-search databases with Walsh-generated queries (Sec. III.A, Fig. 1) and to compressive/random-coding databases (Sec. III.B). The paper further notes that Grover’s Hamming-weight-1 search does not admit a similar single-query violation of the classical bound, because the classical bound is known to be tight for that problem.

2 Claims Table

ID	Claim	Type	Testable?	Testable?
C1	For any $y \in \{0,1\}^n$, a single Bernstein–Vazirani-style quantum query returns y with probability 1 (Sec. II, Eq. (2.6)).	unitary + measurement	yes (small n)	yes
C2	Classical determination of y requires at least $M \geq n/\log_2(n+1)$ queries (Eq. (2.1)).	information-theoretic bound	yes (counting)	yes
C3	The Walsh-circuit binary-search variant (Sec. III.A, Eq. (3.6)) yields mutually orthogonal states $\{ \psi_y\rangle\}$, so a single query plus measurement identifies y .	unitary + measurement	yes (small n)	par
C4	The random-coding variant (Sec. III.B) recovers y with high probability using $m = \log_A k + \ell$ random queries.	probabilistic combinatorics	yes (Monte Carlo)	no
C5	Grover’s Hamming-weight-1 search cannot beat the classical information-theoretic bound (background statement, Refs. [9,10]).	lower-bound literature reference	no (external result)	N/A

3 Method

- Downloaded <https://arxiv.org/pdf/quant-ph/9705041> into `work/paper.pdf` and generated plain-text via `pdftotext` for both layout-preserving and reading-order variants (used as `extraction/marker.md` and `extraction/nougat.mmd` fallbacks; see the extraction files for provenance notes).
- Created a Python 3 virtual environment at `.venv` and installed `qiskit==2.5.0`, `qiskit-aer==0.17.2`, `numpy==2.5.1`.
- Implemented the BV single-query circuit (`report/evidence/bv_single_query.py`):
 - Query register X of $n = 4$ qubits + phase-kickback ancilla B prepared in $|-\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$.
 - Hadamards on X to build the uniform superposition.
 - Oracle U_y implemented as CNOTs from x_i to b for every i with $y_i = 1$, realising $|x\rangle|b\rangle \rightarrow |x\rangle|b \oplus (x \cdot y)\rangle$.
 - Final Hadamards on X , then computational-basis measurement.
- Enumerated all $2^n = 16$ possible databases $y \in \{0,1\}^4$, running each in both (a) exact statevector mode (`Statevector.from_instruction`) and (b) 4096-shot sampling on `AerSimulator`. Recorded exact success probability and shot-count success probability.
- Ran the coin-weighing variant (`report/evidence/coin_weighing.py`): $n = 4$ coins with a single defective coin (all 4 Hamming-weight-1 databases) and, as a scaling check, $n = 8$ over all $2^8 = 256$ databases.
- Computed comparison baselines analytically:
 - Classical single-query success on random guess: $1/2^n = 1/16$.
 - Grover 1-iteration for $N = 2^n = 16$, 1 marked item: $P = \sin^2(3\theta)$ with $\theta = \arcsin(1/4)$, giving $P \approx 0.4727$.

- Grover 1-iteration for $N = 4$, 1 marked: analytic $P = \sin^2(3\pi/6) = 1$ (exact); textbook approximation $(3/4)^2 = 0.5625$.

3.1 Exact commands

```
python3 -m venv .venv
source .venv/bin/activate
pip install qiskit qiskit-aer numpy
python3 report/evidence/bv_single_query.py # 16-database sweep, n=4
python3 report/evidence/coin_weighing.py # coin-weighing n=4 (HW1) + n=8 (all 256)
```

4 Results vs Paper

4.1 C1: single-query recovery of y (main claim), $n = 4$

For all 16 databases $y \in \{0, 1\}^4$, the exact statevector success probability was $P_{\text{exact}} = 1.0$ and the 4096-shot sampled success probability was $P_{\text{shots}} = 1.0$ (perfect deterministic recovery). Full per- y table in `report/evidence/bv_results.json`.

Quantity	Paper ($n=4$)	This replication	Verdict
$P(\text{recover } y \mid 1 \text{ query})$	1.0	1.0 (all 16 y)	MATCH
Classical min queries for certainty	$n / \log_2(n+1) = 1.72 \rightarrow 2$	2 (arithmetic)	MATCH

4.2 Coin-weighing extension, $n = 4$ (single defective) and $n = 8$ (all databases)

Both extensions reproduce $P = 1.0$ across the full database space: $n = 4$ HW-1 (4 databases) all recover perfectly; $n = 8$ (256 databases) all recover perfectly. See `report/evidence/coin_weighing_results.json`.

4.3 Grover baseline for context (task-brief comparison)

The task brief asked for a comparison to Grover 1-iteration. This paper does *not* claim to beat Grover on Hamming-weight-1 search; it claims the opposite (Sec. I, para. 5 of the intro): Grover’s single-marked-item search is provably at the classical bound, and the paper’s speedup applies instead to the *parity/coin-weighing* class. For completeness we recorded the 1-iteration Grover analytic success rate for $N = 16$ ($P \approx 0.4727$) and $N = 4$ ($P = 1$ analytic; 0.5625 textbook). The paper’s single-query $P = 1$ result therefore *coexists* with, and does not contradict, Grover’s optimality on the Hamming-weight-1 problem.

5 Per-Claim: What Worked / What Didn’t

C1 (main). Reproduced perfectly on Qiskit Aer statevector, all $2^4 = 16$ databases, exact $P = 1$. **WORKED.**

C2 (classical bound). Elementary arithmetic; matches. **WORKED.**

C3 (Walsh binary-search). Structurally identical to C1 for the uniform-probability case (the paper itself notes “The Huffman construction results in Walsh queries in the equal-probability-case”). Not implemented as a separate variant. **PARTIAL** — subsumed by C1.

C4 (random coding). Not implemented in this pass. **NOT TESTED.**

C5 (Grover lower-bound remark). External literature statement, not a numerical claim to be reproduced. **N/A.**

6 Verdict and Justification

Verdict: REPLICATED (headline single-query success probability of $P = 1$ for the Bernstein–Vazirani/Terhal–Smolin single-query database protocol, reproduced exactly by Qiskit Aer statevector simulation across every database in the tested spaces $\{0, 1\}^4$ and $\{0, 1\}^8$).

Justification: the paper’s central testable numerical claim in Sec. II is $P_{\text{success}} = 1$ for arbitrary y after a single quantum query. Our simulation reproduces this exactly for 16 ($n=4$) and 256 ($n=8$) databases, with no shot-noise deviations even at 1024–8192 shots. The classical information-theoretic minimum ($n/\log_2(n+1)$) is trivially reproduced by arithmetic. Two of the paper’s five identified claims are structurally subsumed by the main claim (C3) or external literature (C5); one (C4, random-coding) was not attempted this pass (see §??).

7 Open Questions

The five open questions grounded in this replication are enumerated in `report/open_questions.json`. Summarised:

- Q1** How does the single-query success probability degrade under a realistic noise model (single- and two-qubit depolarising noise, amplitude damping) as n grows? At what per-gate error rate does the BV single-query advantage cross the classical break-even?
- Q2** For non-uniform database prior $\{p_y\}$, does the paper’s Huffman construction actually retain single-query recovery on all sufficiently likely codewords when the code lengths ℓ_i vary? We saw the paper handle this only informally — a systematic quantitative analysis on a fixed non-uniform distribution is missing.
- Q3** The paper conflates two cost models: gate count (fine on an idealised architecture) and depth (potentially $n \log n$ for the Walsh circuit). What is the actual asymptotic advantage under a fixed physical error budget, treating T-count / depth as the resource?
- Q4** Sec. III.B (random-coding) predicts a collision probability $p_{\text{col}} = 1 - (1 - A^{-m})^{k-1}$. We did not simulate this. A direct Monte-Carlo comparison of empirical vs analytic collision rates for ($A = 2$, $n = 6$, $k \in \{4, 8, 16\}$) would test whether the paper’s continuum estimate holds at small n .
- Q5** The paper’s “single query” assumes coherent access to the oracle. What happens under partial coherence (dephasing of the query register between preparation and oracle call) — does success degrade smoothly, or is there a coherence-threshold behaviour analogous to fault-tolerance thresholds?

8 Provenance

- Paper PDF: `paper.pdf` (arXiv:quant-ph/9705041)

- Extractions: `extraction/marker.md`, `extraction/nougat.mmd`
- Simulation code: `report/evidence/bv_single_query.py`, `report/evidence/coin_weighing.py`
- Raw results: `report/evidence/bv_results.json`, `report/evidence/coin_weighing_results.json`
- Tools: Python 3.14, Qiskit 2.5.0, Qiskit-Aer 0.17.2, NumPy 2.5.1
- Host: CherryRd (Darwin 25.3.0, x64)